

Newton-Raphson for Nonlinear Regression (Michaelis-Menten) — Exam-Ready Short Answers

Q1. What is being fitted, and what's the goal?

Model: $y = \theta_1 x / (\theta_2 + x) + \varepsilon$ (Michaelis-Menten). Goal: $\min \text{RSS } S(\theta) = \sum (y_i - \theta_1 x_i / (\theta_2 + x_i))^2$ via Newton-Raphson (nonlinear $\rightarrow$ no closed form).

Q2. Data generation?

$n=50$, $x=1:50$, true $\theta=(5,2)$, $e \sim N(0,0.2^2)$; $y = 5x/(2+x) + e$.

Q3. Gradient (score) components?

$\partial S / \partial \theta_1 = -2 \sum r_i \cdot x_i / (\theta_2 + x_i)$; $\partial S / \partial \theta_2 = 2 \sum r_i \cdot x_i / (\theta_2 + x_i)^2$, where $r_i = y_i - \hat{y}_i$.

Q4. Hessian entries (h11,h12,h21,h22)?

$h11 = 2 \sum (x_i / (\theta_2 + x_i))^2$; $h12 = h21 = 2 \sum [x_i / (\theta_2 + x_i)^2] [y_i - 2\theta_1 x_i / (\theta_2 + x_i)]$;

$h22 = -2 \sum [\theta_1 x_i / (\theta_2 + x_i)^3] [2y_i - 3\theta_1 x_i / (\theta_2 + x_i)]$.

Q5. Matrix Newton-Raphson update?

$\theta^{(k)} = \theta^{(k-1)} - H^{-1} \cdot \text{grad}(S)$; in R: `solve(H) %*% H1` ($H1 = \text{gradient vector}$). Division $\rightarrow$ matrix inverse.

Q6. Why start theta at (5,2) — issue?

(5,2) is truth used to simulate y , also used as NR start — tests update mechanics, not robustness to bad starting guess.

Q7. Convergence check?

if $\max(|\theta^{(k)} - \theta^{(k-1)}|) < \varepsilon = 10^{-6} \rightarrow$ break (both params must barely move).

Q8. Bug in the loop?

`cat("Converged...")` placed AFTER break $\rightarrow$ unreachable, never prints; doesn't affect final `theta[k,]` result.

Q9. Role of h21 = h12?

Hessian is symmetric (mixed partials equal) $\rightarrow$ reuse h12, avoid recomputing.

Q10. Actual numeric results (seed=123)?

- Converges at $k = 47$ iterations.
- Final step change $\approx 9.9 \times 10^{-7}$ ($<$ tolerance).
- $\theta \approx (4.9866, 1.9148)$; true $\theta = (5, 2)$.

Q11. Why does theta drift from (5,2) then re-approach it?

Start = population truth, but noisy $y \rightarrow$ NR converges to sample least-squares fit, not exact truth; small deviation = noise effect.

Q12. What does the fit plot show?

Data: rising-then-saturating scatter. Blue = NR fit curve, red dashed = true curve (5,2) — nearly overlap $\rightarrow$ confirms correct convergence.

Q13. Conclusion?

Extends 1-D NR (score/Hessian scalars) to 2-D (gradient vector/Hessian matrix); `solve()` + `%*%` replace division; converges accurately to LS estimate.